Week 5 Bonus Quiz

Functions that work on lists, without `LinearAlgebra`

Both questions are optional bonus questions. The grading environment has no `LinearAlgebra`, so `dot` and `norm` are unavailable: write the loop yourself.

Question 2 — normalized inner product

Write a function that receives two lists of numbers and returns their inner product, normalized by their lengths:

$$\frac{a \cdot b}{|a|\,|b|}.$$

Signature:

```
function norm_inner_product(v1, v2)
    # ...
end
```

Return value: a number. You may assume the input is two lists containing only `Int` or `Float64` values.

- If the lists have different lengths, return `nothing`.
- If the lists are empty, return `0`.

A zero vector makes the normalized product undefined; the tests do not include that case.

Example: for `v1 = [3.0, 4.0]` and `v2 = [6.0, 8.0]` the result is `1.0`. For `[1.0, 0.0]` and `[0.0, 1.0]` it is `0.0`.

Question 4 — counting orthogonal pairs

Two vectors are orthogonal when their inner product is zero. Write a function that receives a list of vectors (a list of lists) and returns the number of pairs of vectors that are orthogonal to each other:

```
function num_of_orthogonal(vectors)
    # ...
end
```

Return value: the number of orthogonal pairs in the list.

```
num_of_orthogonal([[1, 0, 0], [0, 1, 0], [0, 0, 1]])              # 3
num_of_orthogonal([[0, 0, 0], [0, 1, 0], [0, 0, 1]])              # 3
num_of_orthogonal([[0, 0], [1, 2], [10, 5]])                      # 2
num_of_orthogonal([[1, 1, 1, 1], [2, 1, 3, 3], [0, 0, 100, 33],
                   [8, 8, 8, 1.5], [9, 9, 9, 9]])                 # 0
```

You may assume that the input is a list, that the inner lists have equal length, and that they contain only numbers. Count each pair once: the pair (i, j) and the pair (j, i) are the same pair, and a vector is not paired with itself. Avoid duplicated code; use one inner-product helper for both questions.

Call	Result
`norm_inner_product([1.0, 0.0], [-1.0, 0.0])`	`-1.0`
`norm_inner_product([1, 2, 3], [1, 2])`	`nothing`
`norm_inner_product(Float64[], Float64[])`	`0`
`num_of_orthogonal` on the 4D basis	`6`